

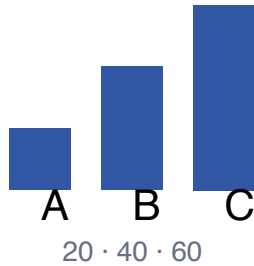
Choosing Effective Visual Encodings

From possible encodings to good encodings

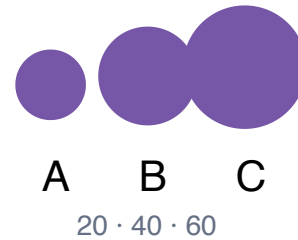
Information Visualization · Lecture 4

One attribute can be encoded in many ways

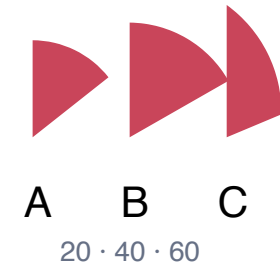
Length



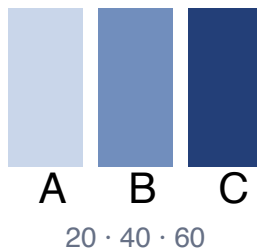
Area



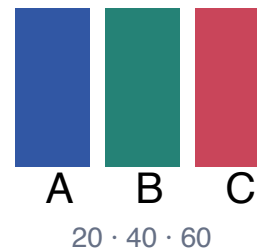
Angle



Luminance



Hue



**All five encodings
distinguish the three values.
Are they equally good?**

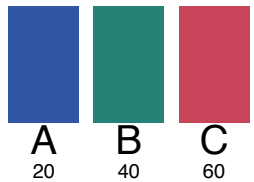
Agenda

-
- | | | |
|----|---|--|
| 01 | Expressiveness and effectiveness | Choose encodings that preserve meaning and support the task. |
| 02 | Channel discriminability and compatibility | Check channel capacity, separability, and interference. |
| 03 | Redundant encoding and pop-out | Reinforce important signals and guide attention. |
-

A good encoding must pass two tests

Expressiveness

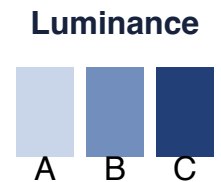
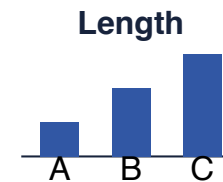
Does the encoding represent the meaning and relevant relationships of the data correctly?



The hues distinguish A, B, and C, but do they show that **20 < 40 < 60**?

Effectiveness

Does the encoding make the information important to the task easy and accurate to perceive?



Which encoding makes it easier to estimate **how much** A, B, and C differ?

A visualization should first be **expressive**, then as **effective as possible for the task**.

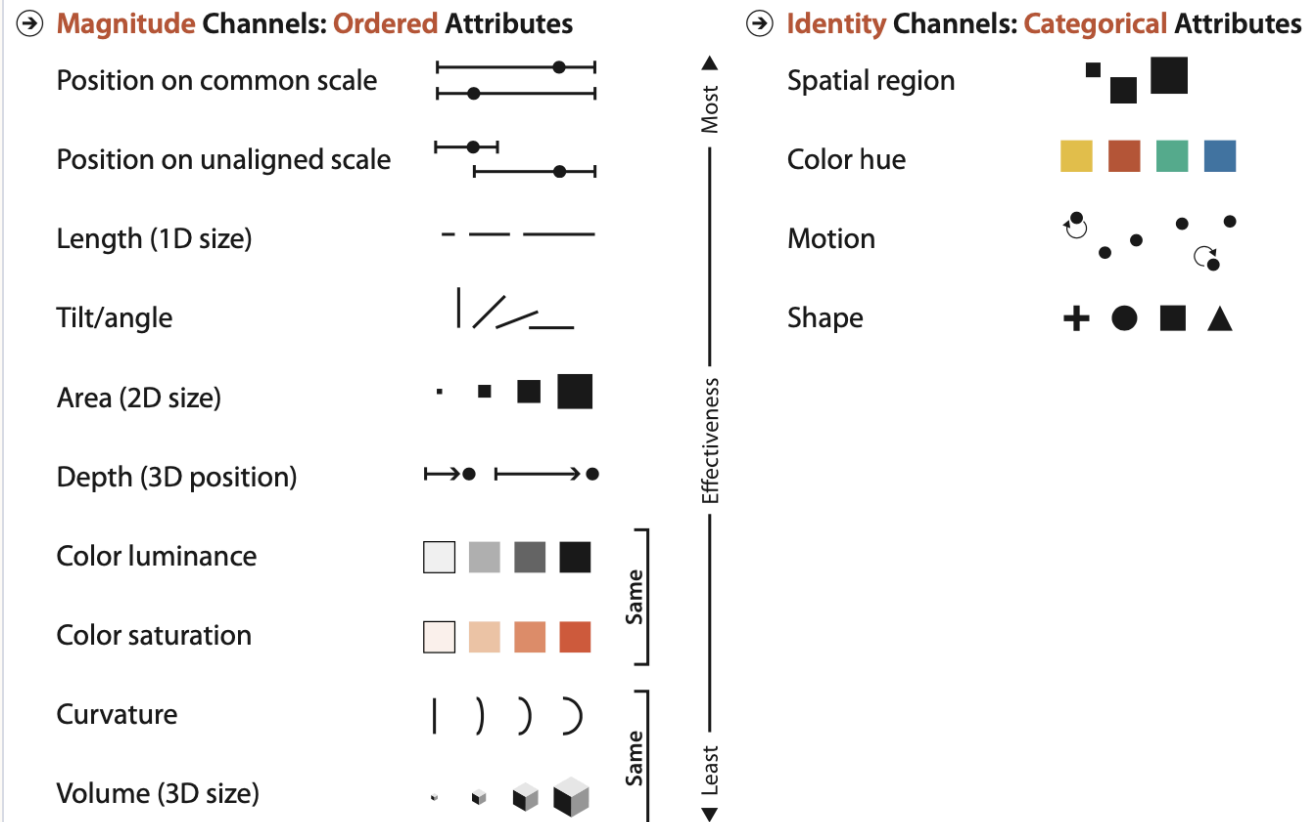
Expressiveness preserves the meaning of the data

The expressiveness principle requires the meaning implied by a visual encoding to match the meaning of the data. The encoding must preserve the relationships needed for the task and must not suggest relationships that the data does not contain.

- **Categorical attributes** need channels that show identity or difference without implying order or magnitude.
- **Ordered attributes** need channels that viewers perceive as ordered.

Design check: Does the visual imply the same relationships as the data?

Identity and magnitude channels support different judgements



What is it?

Identity channels

Primarily support judgments of **sameness and difference**.

Hue

Shape

Spatial region

How much?

Magnitude channels

Support judgments of **order or quantitative amount**.

Position

Length

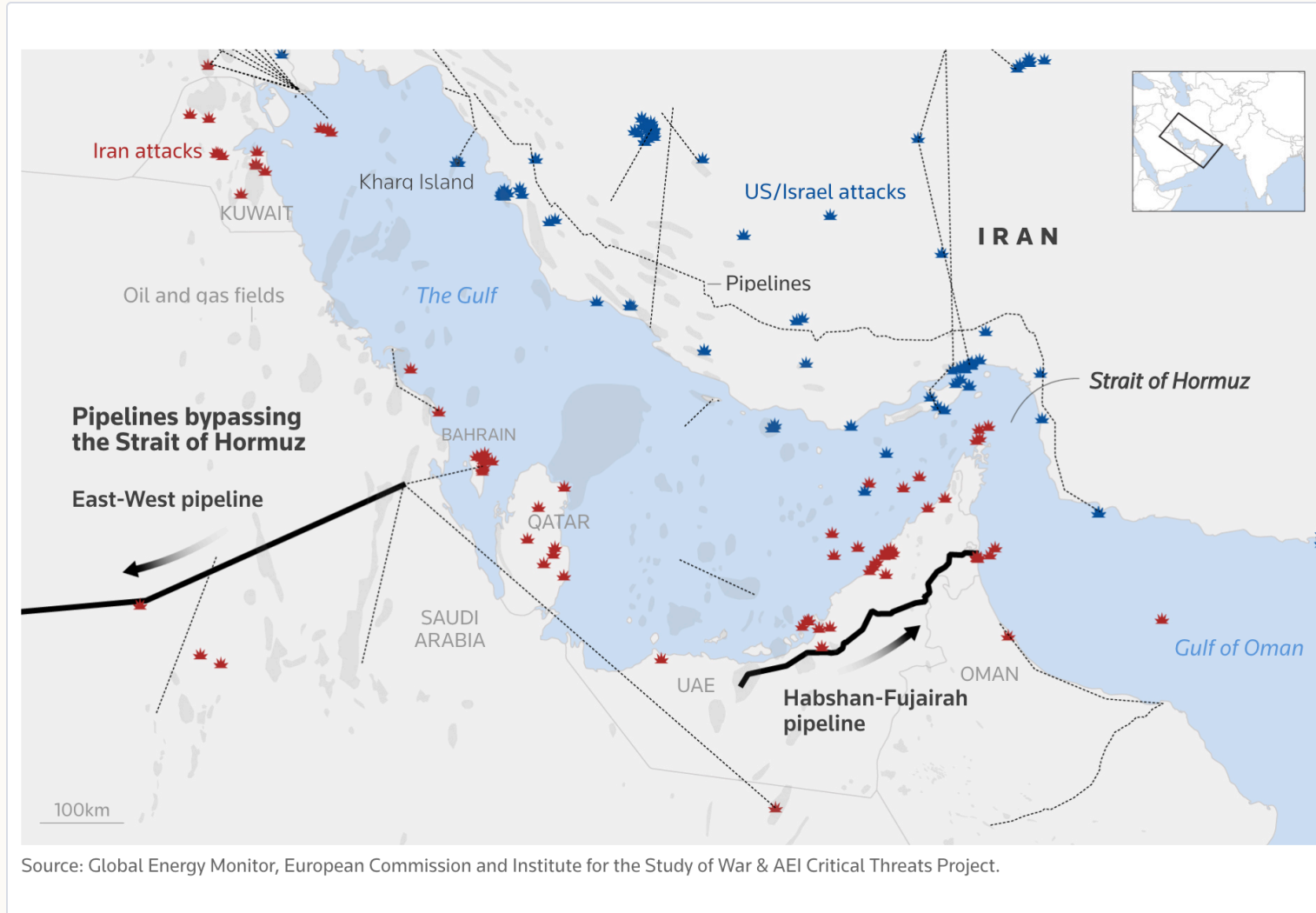
Angle

Area

Luminance

These are primary perceptual uses—not perfectly exclusive categories.

Categorical attributes need identity judgments



Iran, or, US → hue

The viewer needs to determine whether two points belong to the same or different categories.

No order or magnitude comparison is required.

Unordered hue cannot communicate quantitative order

Which is larger?

Is green higher than blue?
Hue provides difference, but
no natural magnitude order.

Different $\neq$ more or less

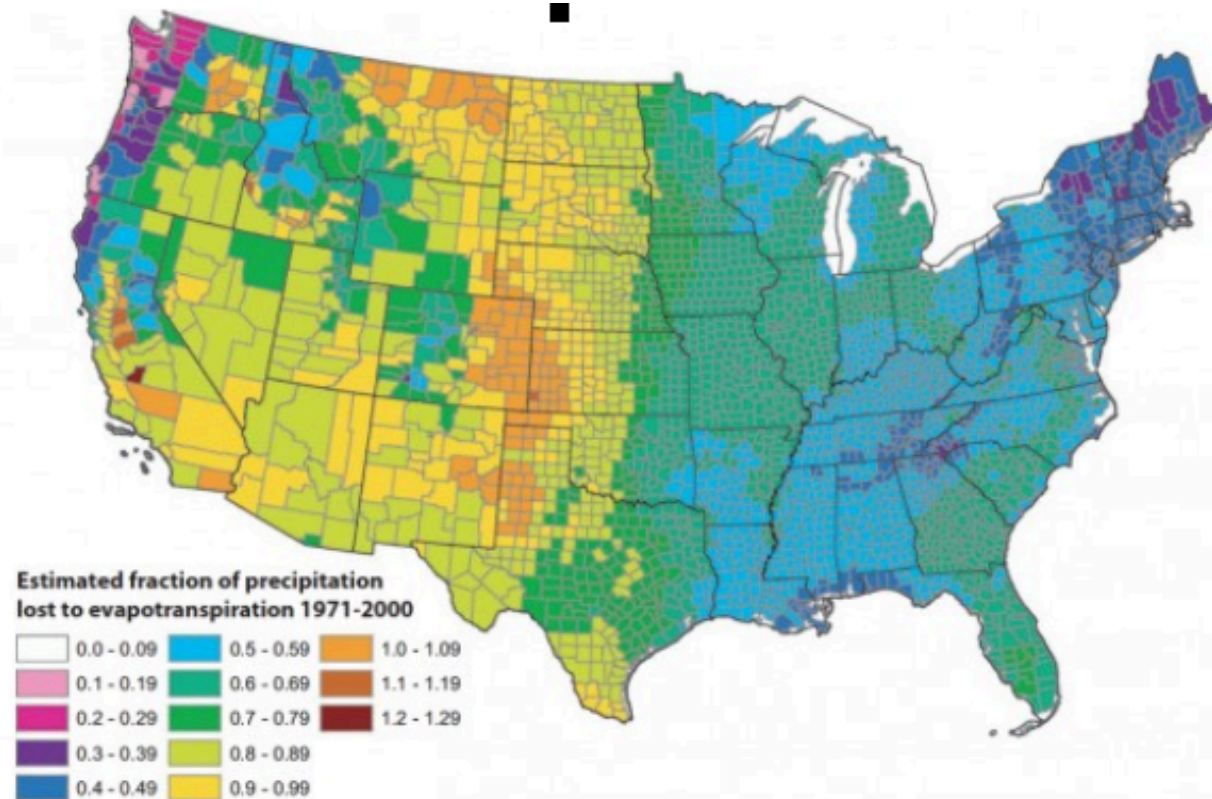
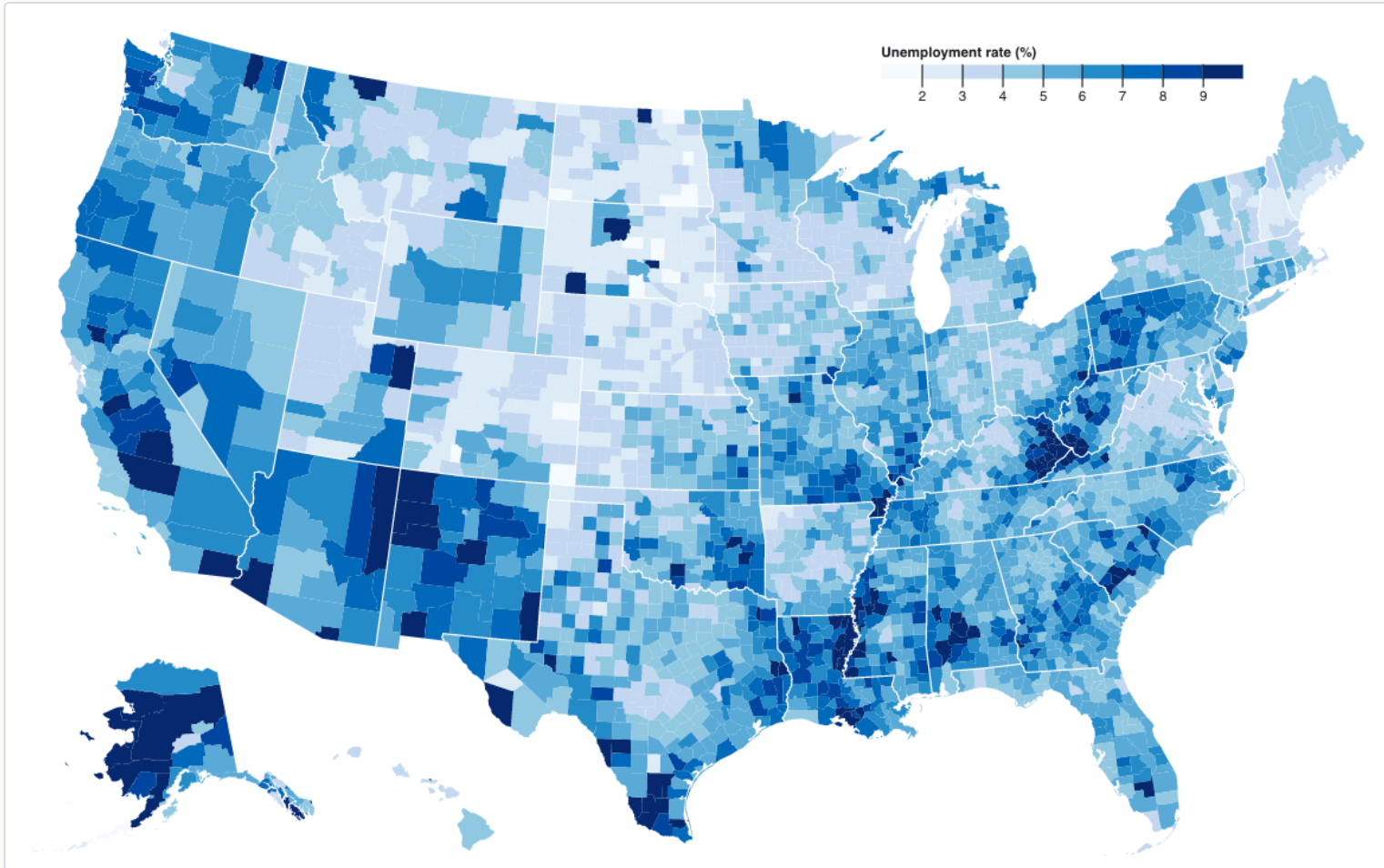


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

© 2000

Ordered lightness can communicate quantitative magnitude



Light → dark

A sequential color scale uses ordered lightness or luminance to represent increasing magnitude.

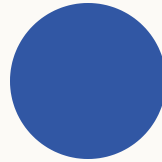
Viewers can recover a visual order even when exact values require the legend.

Size can invent an order that categories do not have

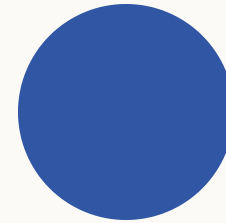
Task: Identify which company each mark represents.



Company A



Company B



Company C

What the data means

A, B, and C are categories. They have no inherent order or magnitude.

What size implies

Increasing circle size suggests an order such as $A < B < C$.

Design check failed: Size introduces a relationship that the data does not contain.

Which mappings are expressive?

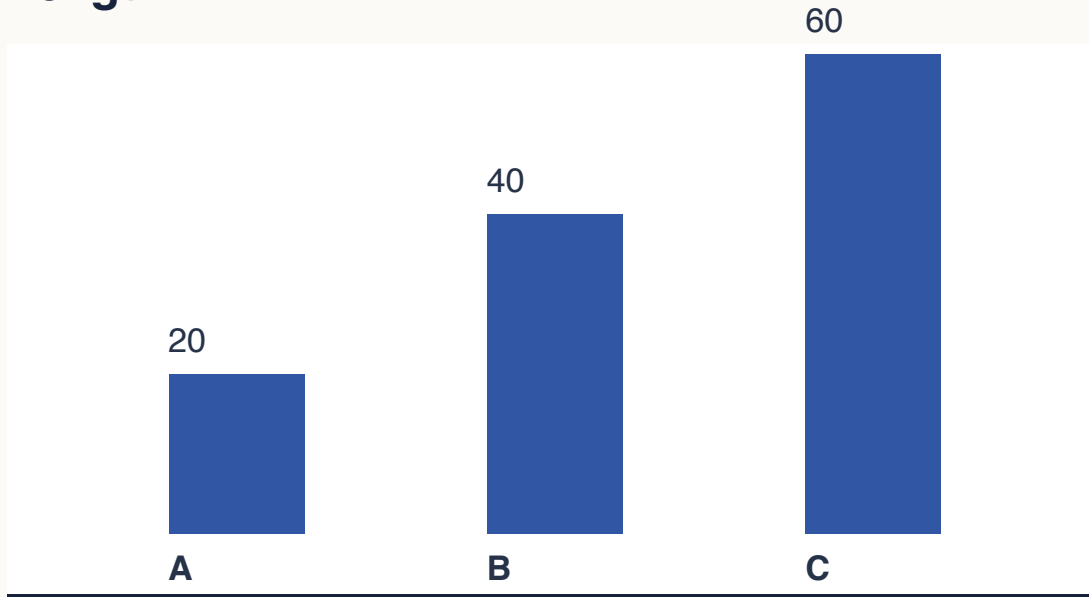
Attribute	Channel	Your judgment
Continent	Hue	Expressive / Problematic
Salary	Hue	Expressive / Problematic
Ranking	Vertical position	Expressive / Problematic
Country	Size	Expressive / Problematic
Temperature	Luminance	Expressive / Problematic
Product category	Shape	Expressive / Problematic

Explain your choice:

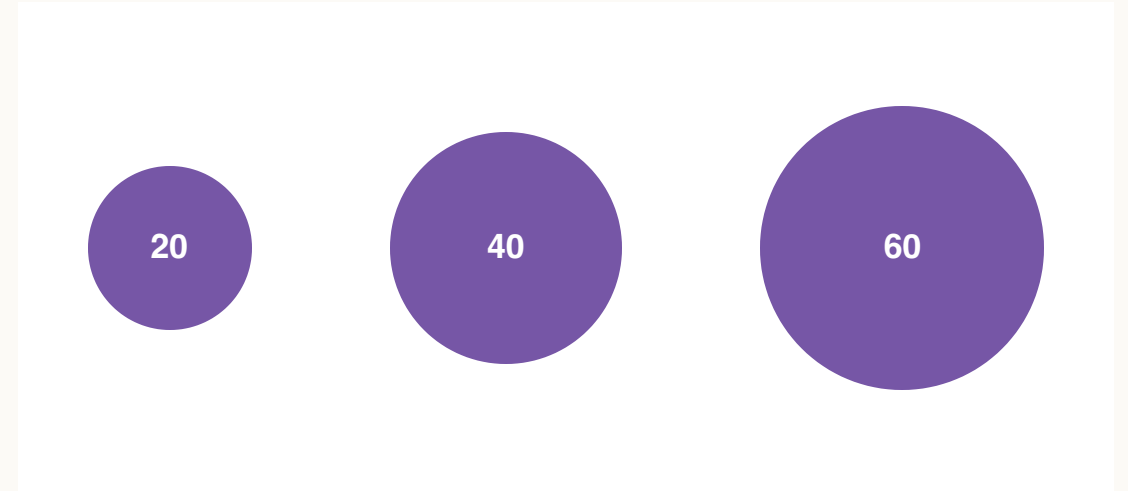
What relationship does the channel support, or falsely imply?

Two expressive encodings can differ in effectiveness

Length



Area



Both preserve magnitude. Which supports more accurate comparison?

Effectiveness assigns strong channels to what matters

The effectiveness principle requires the attributes most important to the visualization task to be encoded using the channels that support the most accurate and efficient perceptual judgments.

Task

What must the viewer judge?



Important attributes

Which data fields determine the answer?



Strong suitable channels

Which channels support that judgment?

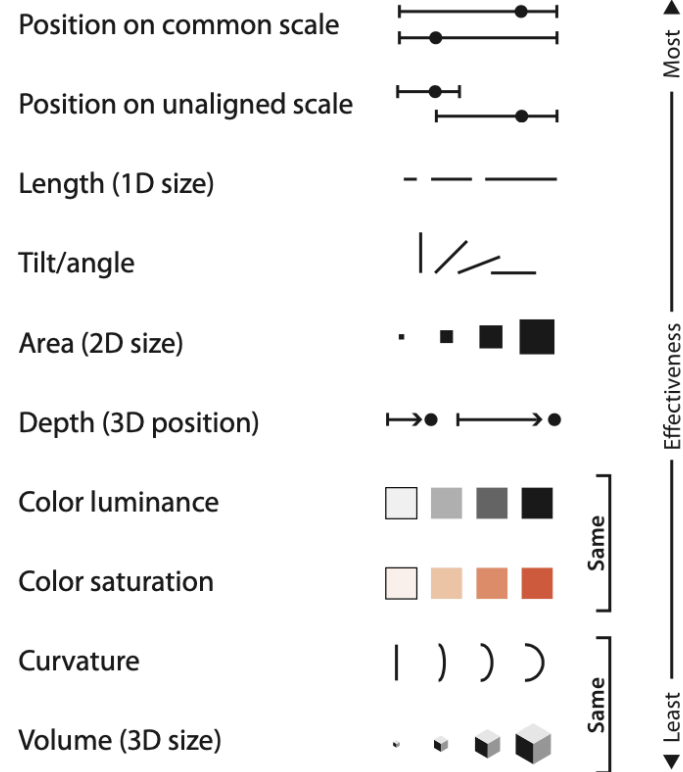
Perceptual accuracy provides evidence for effectiveness

Perceptual accuracy describes how closely a viewer's visual judgment corresponds to the underlying value or objective measurement represented by the graphic.

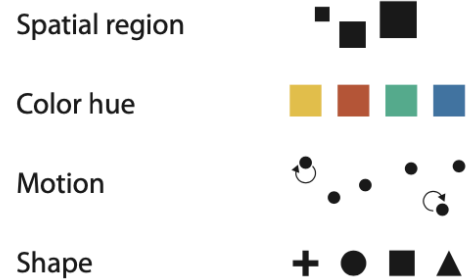
- Researchers compare estimation errors across visual channels.
- A channel that produces smaller errors for a judgment is generally more effective for that judgment.
- Accuracy is not the only design goal, but it is central when viewers must compare quantitative values.

Channels have ranks

➔ Magnitude Channels: Ordered Attributes



➔ Identity Channels: Categorical Attributes



How easily can categories be distinguished?

Ranking of identity channels

Spatial region > Hue > Motion > Shape

How accurately can values be compared?

Ranking of magnitude channels

Selected channels:

Aligned position on a common scale > Unaligned position on identical scales > Length > Angle > Area > Luminance

These rankings summarize general perceptual effectiveness; the best choice still depends on the task and context.

The ranking is guidance—not a command to always use position

TASK A

Compare exact values

Position or length on a common scale strongly improves accuracy.

The ranking matters greatly.

Higher-ranked channels generally support more accurate quantitative comparison; effectiveness remains task-dependent.

TASK B

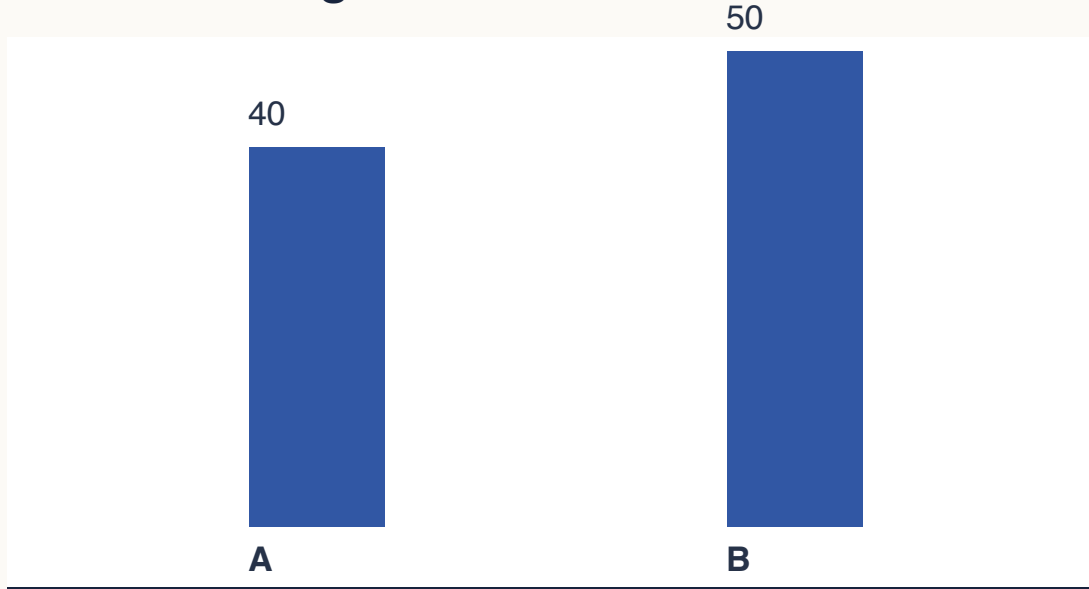
See a rough geographic pattern

Ordered color may be sufficient because spatial context must remain visible.

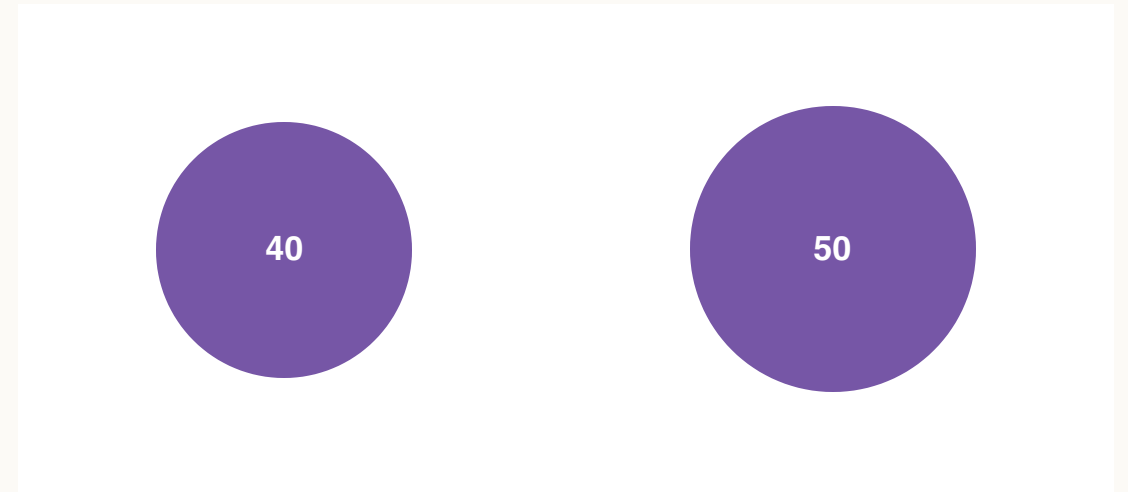
Task and constraints matter.

A common baseline makes 40 versus 50 easier to compare

Bar chart · length



Bubble chart · area



The difference is **perceptual effectiveness**, not expressiveness.

Alignment and a common scale reduce comparison effort

Aligned



The endpoints share a reference, so their difference is directly visible.

Unaligned



The viewer must mentally transport one length before comparing it with the other.

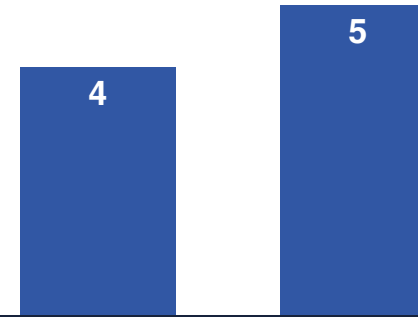
We judge a difference relative to its reference



Shared scale: 0–5

Absolute difference = 1

Relative difference = 100%



Shared scale: 0–5

Absolute difference = 1

Relative difference = 25%

The detectability and meaning of a difference depend on the magnitude of the reference stimulus.

Weber's law formalizes relative sensitivity

Weber's law states that the smallest detectable change in a stimulus is approximately proportional to the magnitude of the reference stimulus. Perceptual judgments therefore depend more strongly on relative differences than on absolute differences.

$$\Delta I / I \approx k$$

The smallest detectable relative change is approximately constant.

I	Reference stimulus intensity
ΔI	Smallest detectable change
k	Approximate constant for a given sensory condition

A small I results in an small ΔI .

Comparison becomes easier when the design supplies context

Shared reference



Common baseline and scale expose the relative difference.

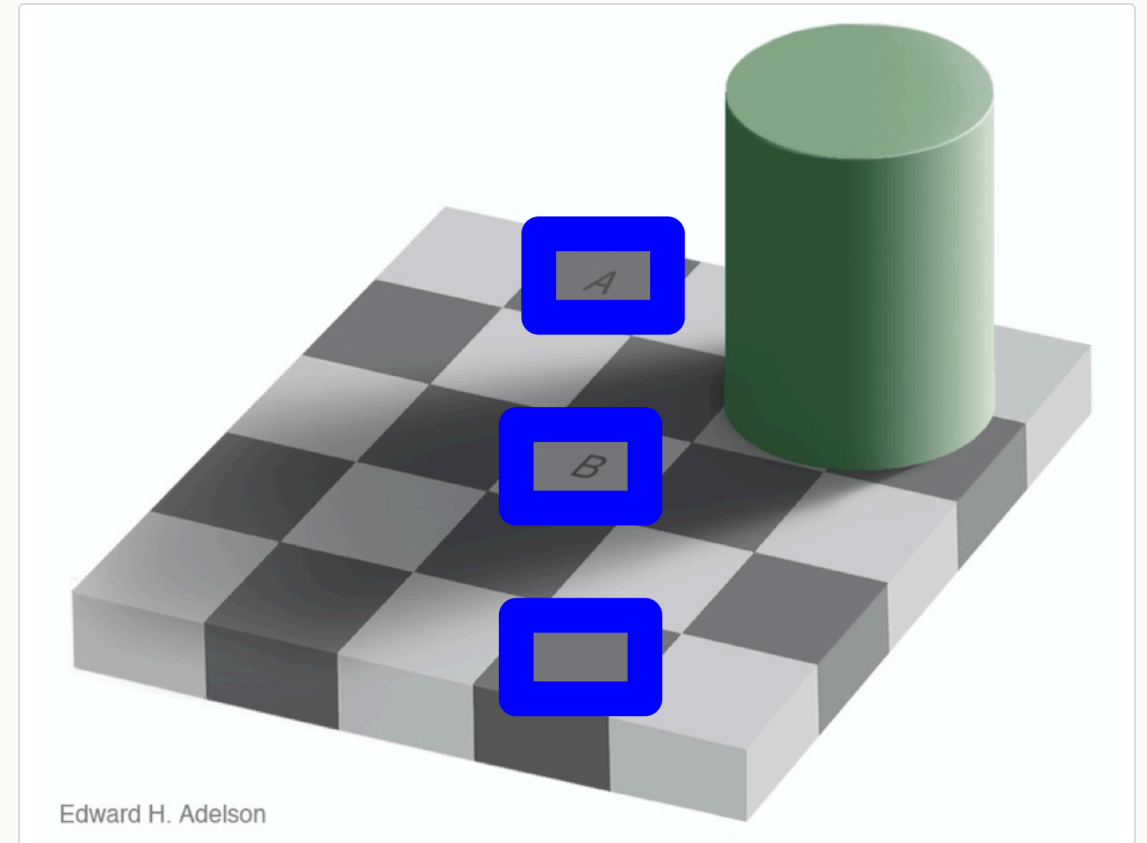
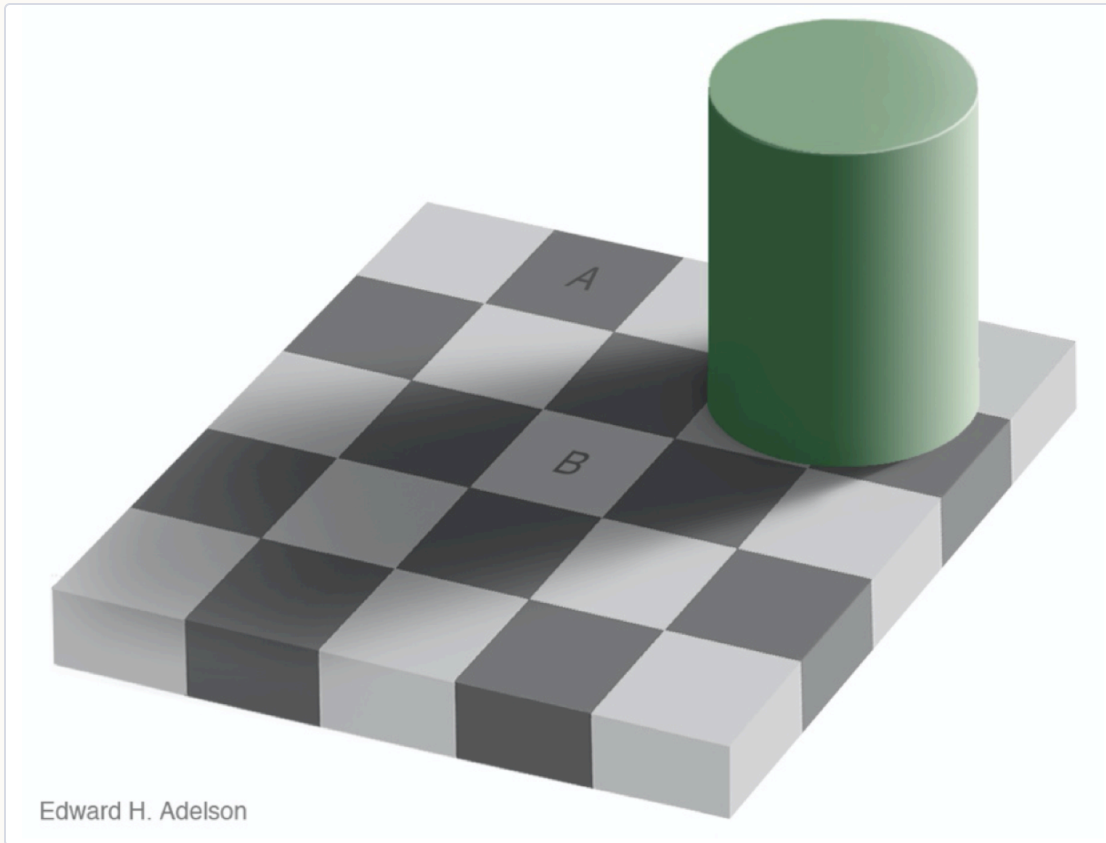
Separate references



Without alignment, viewers must construct the reference mentally.

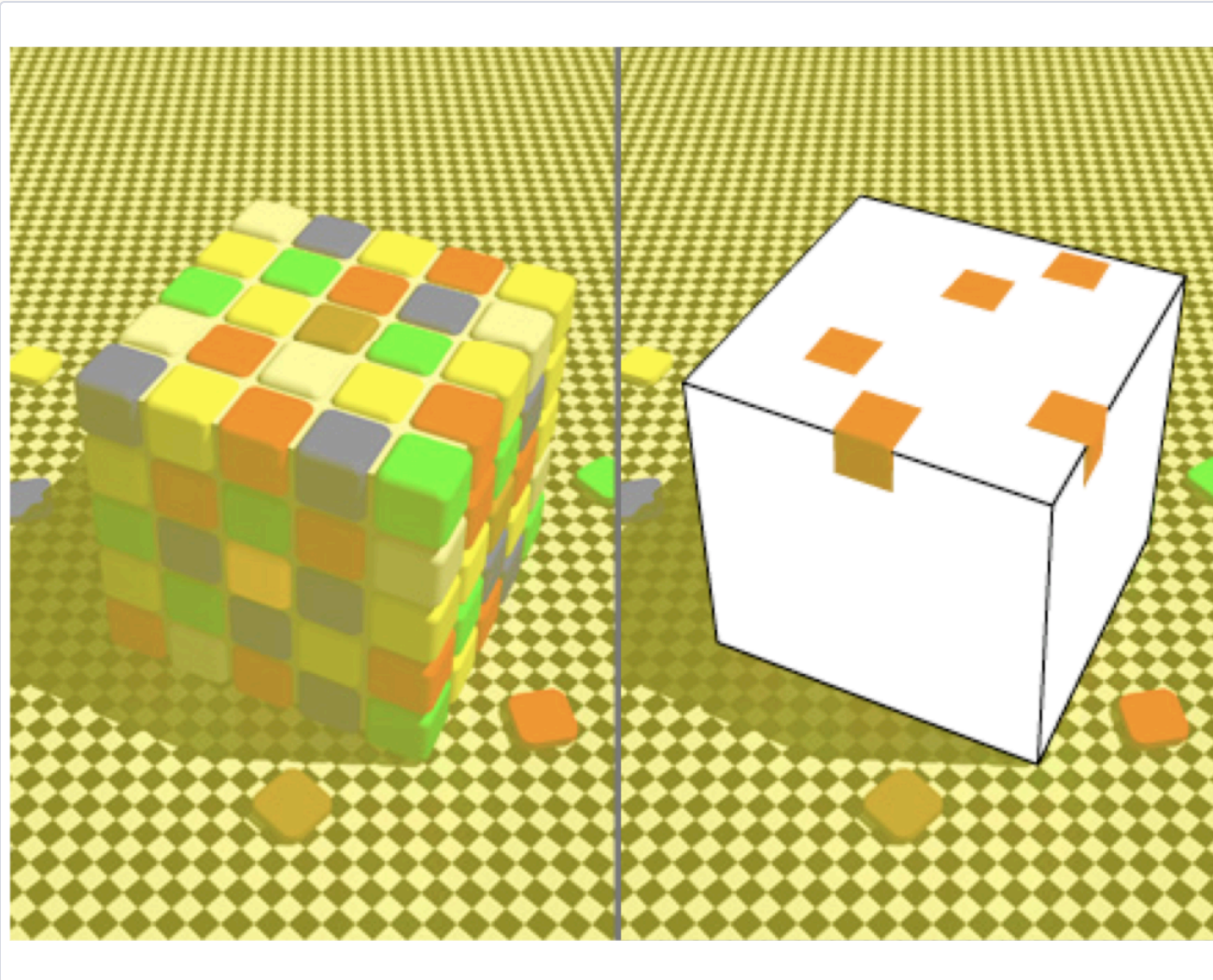
Shared scales, baselines, and reference lines externalize comparison.

The same luminance can look different in different contexts



Square A and B have the **same fill**; their surroundings change their appearance.

Color is contextual—not absolute



What can go wrong

- Equal numeric steps may not produce equal perceptual steps.
- Surrounding colors can change apparent lightness.
- Complex backgrounds make comparison less stable.

Design response

- Use coherent sequential scales for low → high.
- Use diverging scales around a meaningful midpoint.
- Keep backgrounds restrained and check contrast.

A color scale must be judged as part of the complete display, not as isolated swatches.

Discriminability limits how many values a channel can carry

Discriminability is the capacity of a visual channel to represent distinct values that viewers can reliably tell apart. The number and spacing of encoded values must remain within the perceptual capacity of the channel under the viewing conditions and task.

- Three well-separated colors may be easy to name.
- Twenty similar colors may be impossible to distinguish reliably.

Attribute cardinality must fit channel capacity

APPROXIMATELY 6 CATEGORIES

Continent encoded by hue



Well-separated hues can support identity judgments when the marks and legend are easy to inspect.

Often workable

APPROXIMATELY 200 CATEGORIES

Country encoded by hue



A small sample from a much larger set

A unique hue for every country exceeds what viewers can distinguish and remember reliably.

Not reliably discriminable

An encoding may be expressive in theory but unusable in practice.

Capacity depends on palette, mark size, spacing, background, legend lookup, viewing conditions, and task. There is no universal threshold.

Channel capacity depends on the viewing conditions

Hue



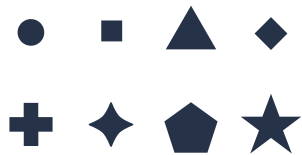
A modest set of well-separated hues can support category identification. Capacity falls when colors are similar, marks are small, or frequent legend lookup is required.

Line width



Only a few clearly separated widths are easy to compare. Fine steps blur together on thin marks and low-resolution displays.

Shape



Simple, clearly different shapes can encode a small category set. Recognition slows when forms become similar or viewers must remember many legend entries.

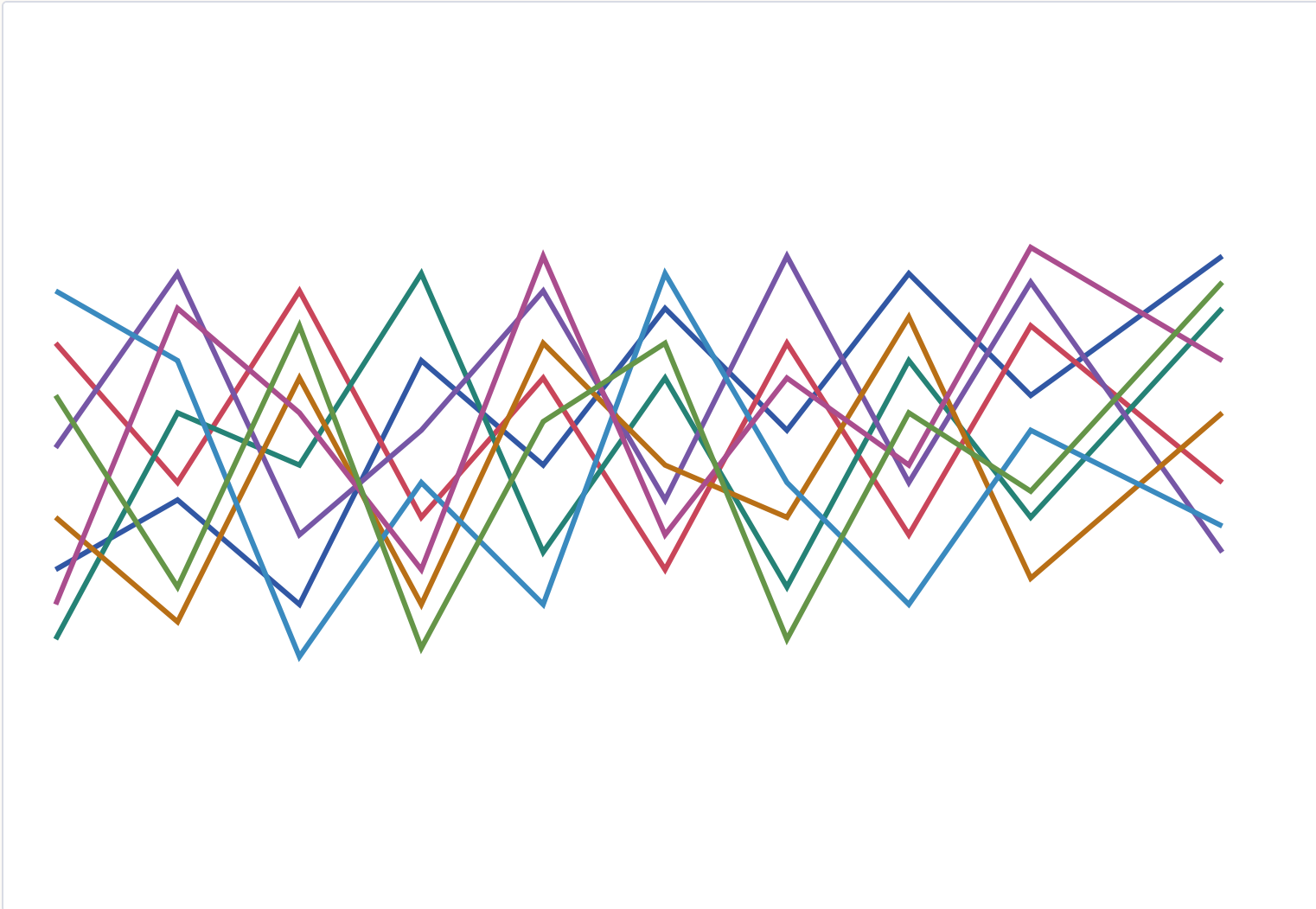
Position



Position supports many ordered values because viewers compare locations along a continuous scale. Resolution still depends on plot size, alignment, and clutter.

No channel has one fixed capacity. Usable resolution depends on the separation between encodings and on how the complete visualization will be viewed and used.

Too many colored lines defeat identification

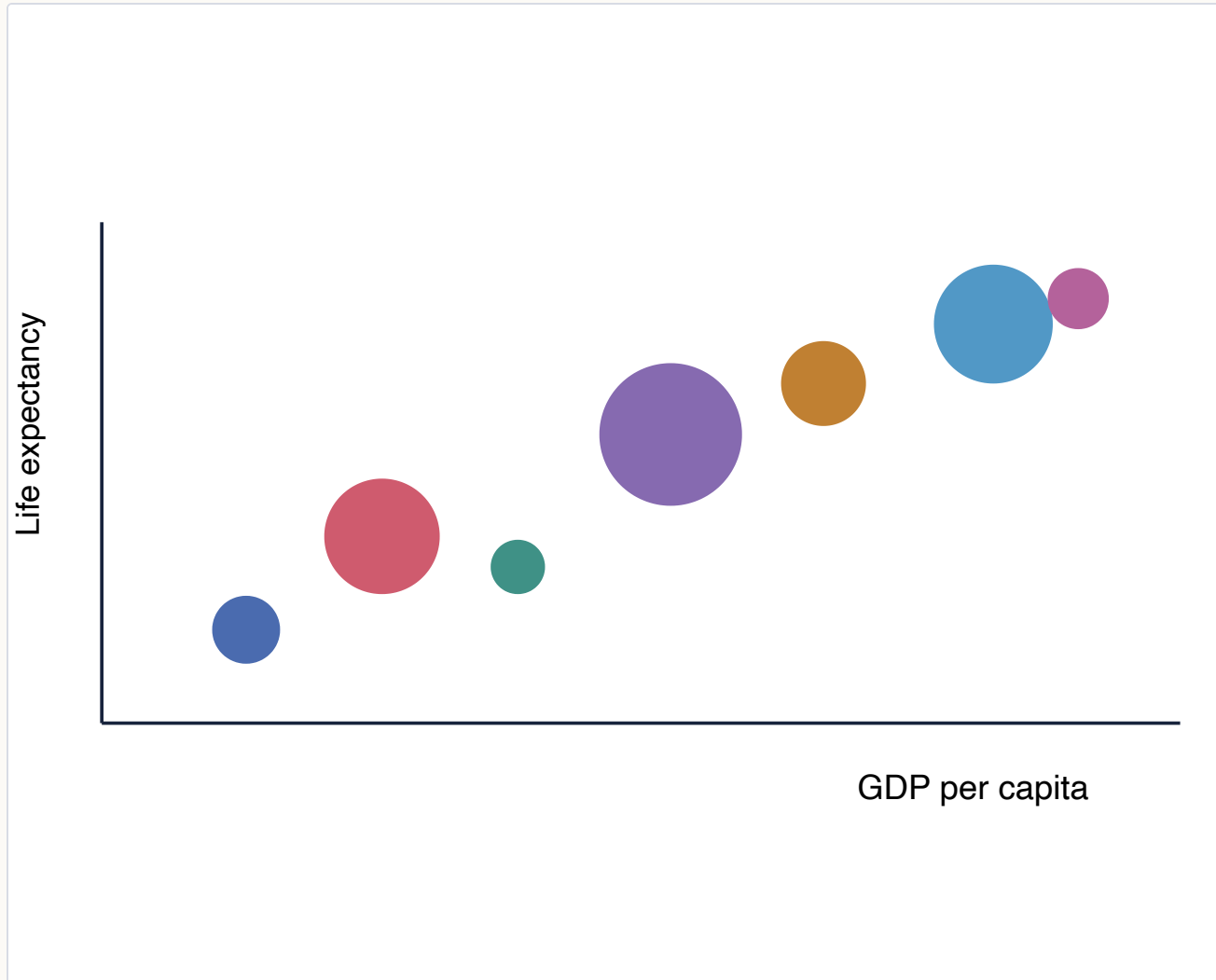


Can you reliably follow one series?

Adding more hues does not solve the identification task.

- Highlight
- Filter
- Group
- Use small multiples

Multiple attributes often share one mark



Four mappings

GDP	x position
-----	------------

Life expectancy	y position
-----------------	------------

Population	size
------------	------

Continent	hue
-----------	-----

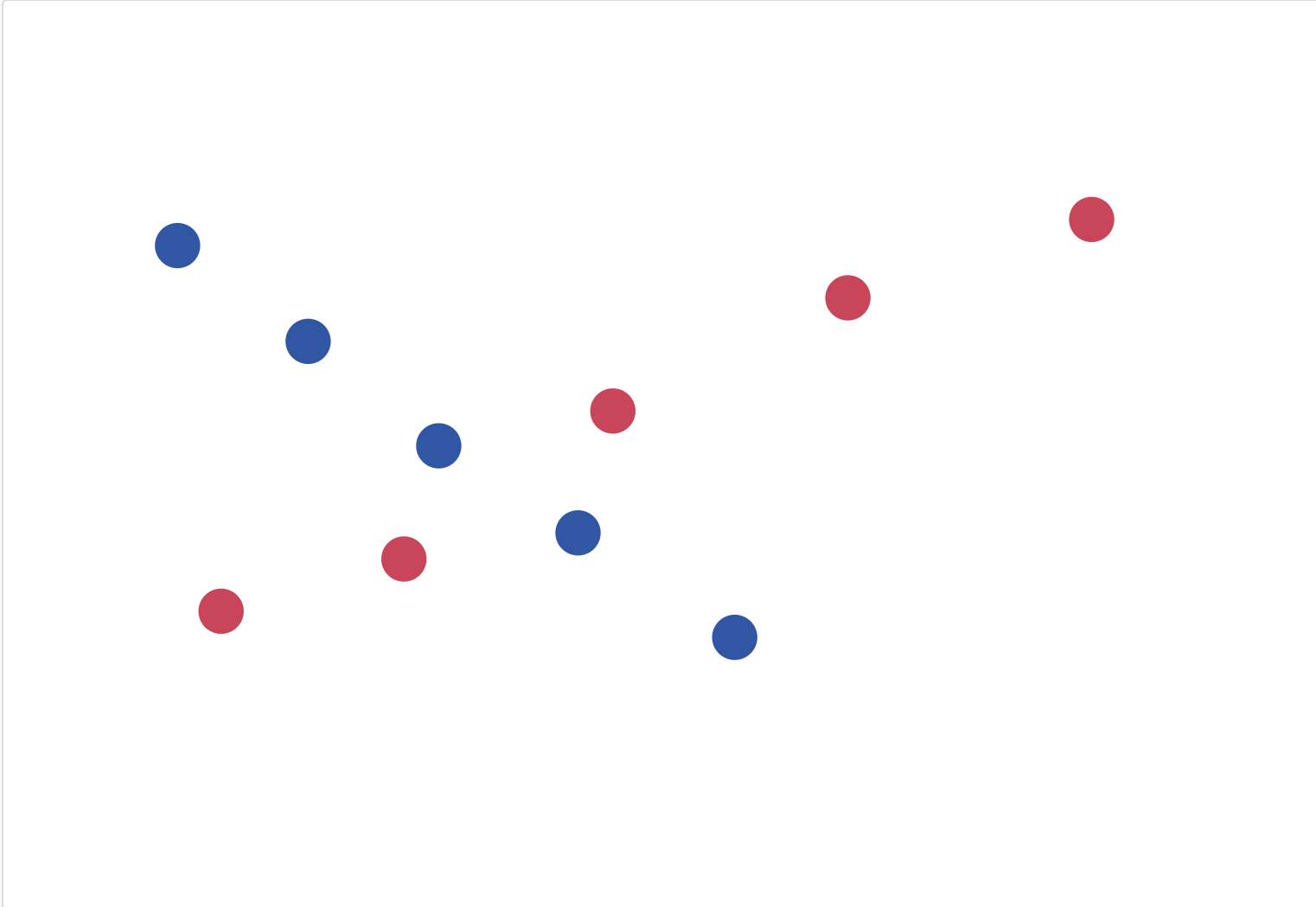
Can viewers read all four independently?

Separability describes interference between channels

Two visual channels are separable when viewers can attend to and interpret the information encoded by one channel without substantial interference from the other.

- Separability is a continuum, not a simple yes/no property.
- It depends on mark design, channel range, contrast, viewing conditions, and task.
- A channel combination that works for one task may fail for another.

Position and hue are often relatively separable



Two judgments remain available

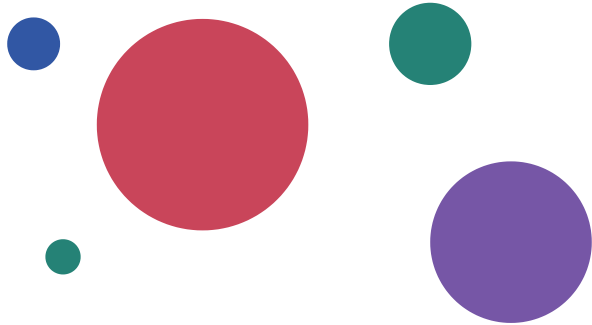
Position reveals the spatial pattern.

Hue reveals category identity.

Viewers can often attend to either with limited interference.

A strong size range can interfere with hue or shape

Size with hue



Large marks attract attention first, so viewers may notice size groups before inspecting the hue categories.

Size with shape



A wide size range can weaken shape recognition because viewers may organize the marks into large and small groups first.

Separability is conditional. Interference increases when one channel creates much stronger visual variation than the other.

Two color mixtures collapse into one perceived color



Attribute A (e.g., GDP) → red intensity

Attribute B (e.g., Population) → blue intensity

Viewers perceive each mixture as a resulting color—not as two independently recoverable hue amounts.

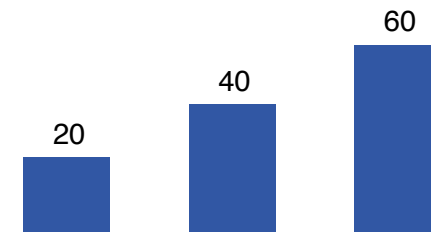
Redundant encoding can strengthen one attribute

Redundant encoding represents the same data attribute using two or more visual channels. It can strengthen an important signal, improve robustness and accessibility, but it also consumes visual capacity and can add unnecessary complexity.

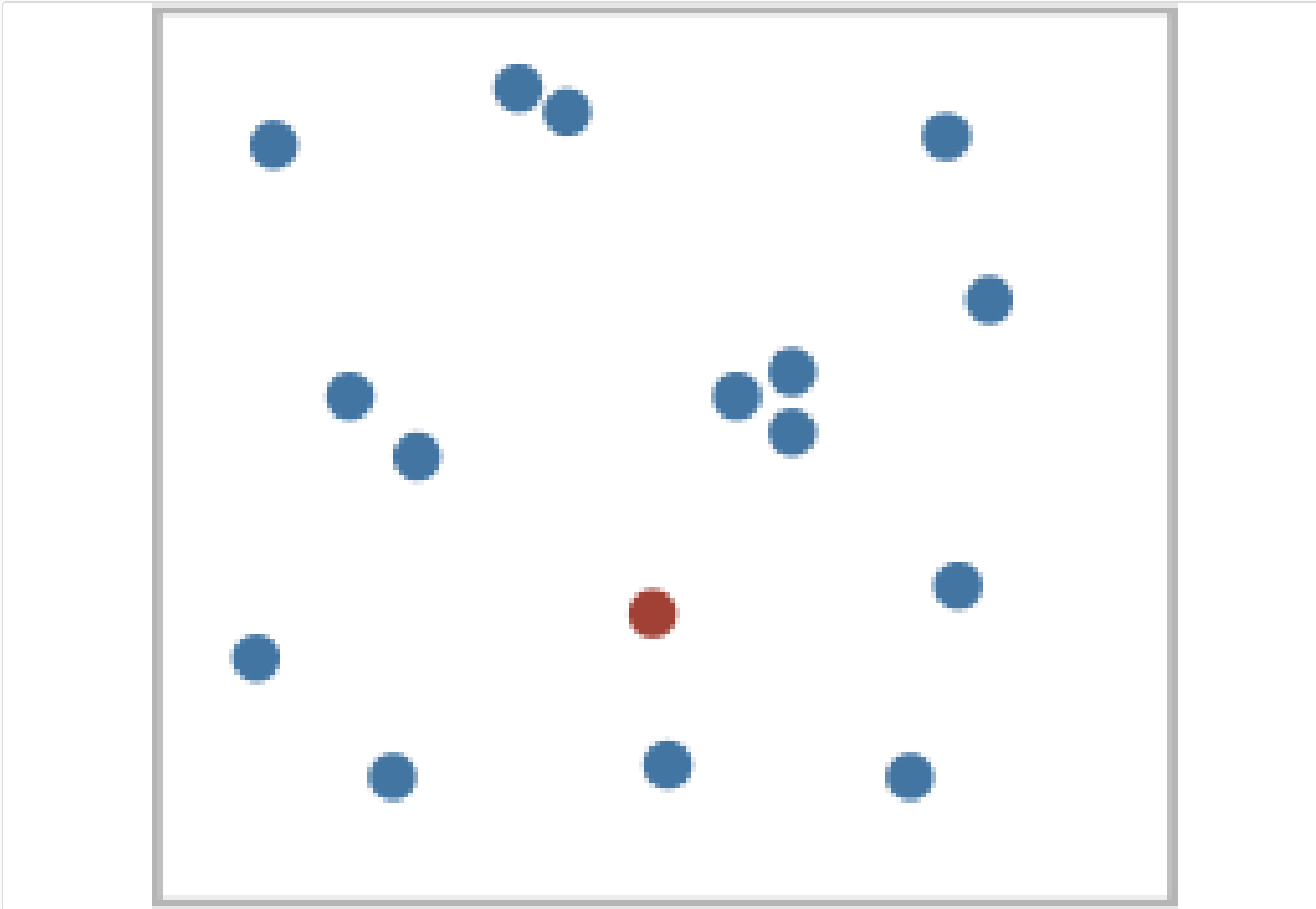
Category → hue + shape



Value → length + label



Pop-out directs attention to a distinctive item

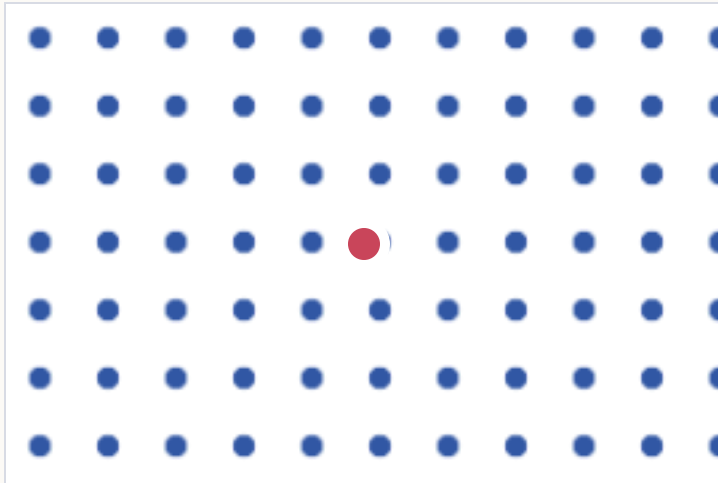


Pop-out

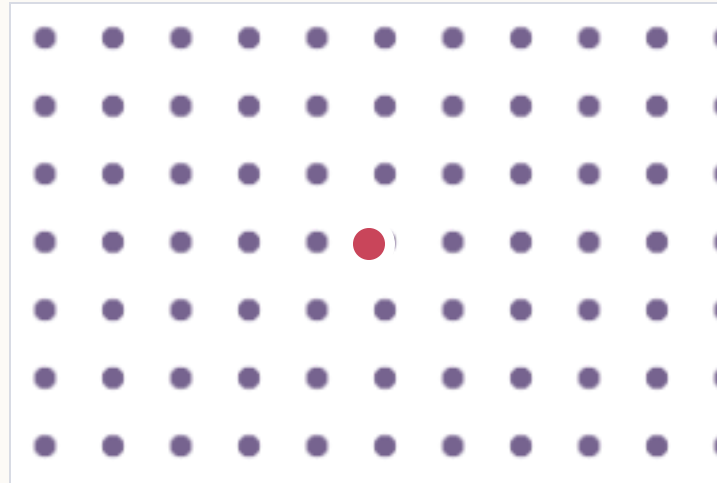
Pop-out occurs when an item differs sufficiently from its surroundings along a salient visual feature and can therefore be detected rapidly without an item-by-item search.

Use it to highlight what matters.

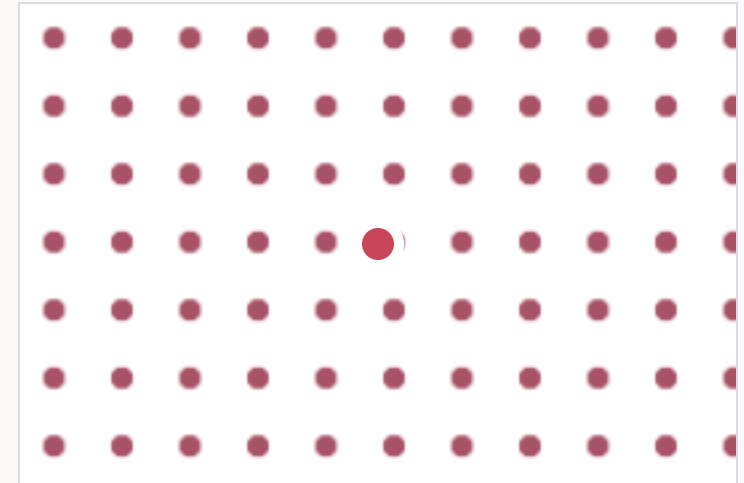
Pop-out weakens as target and distractors become similar



High contrast
rapidly distinctive



Moderate contrast
more search effort



Low contrast
weak salience

Highlighting is a relative design decision: salience depends on the surrounding display.

If everything stands out, nothing stands out

Disciplined emphasis

Reference series

Comparison series

Critical exception

Supporting context

Competing emphasis

Revenue

Costs

Growth

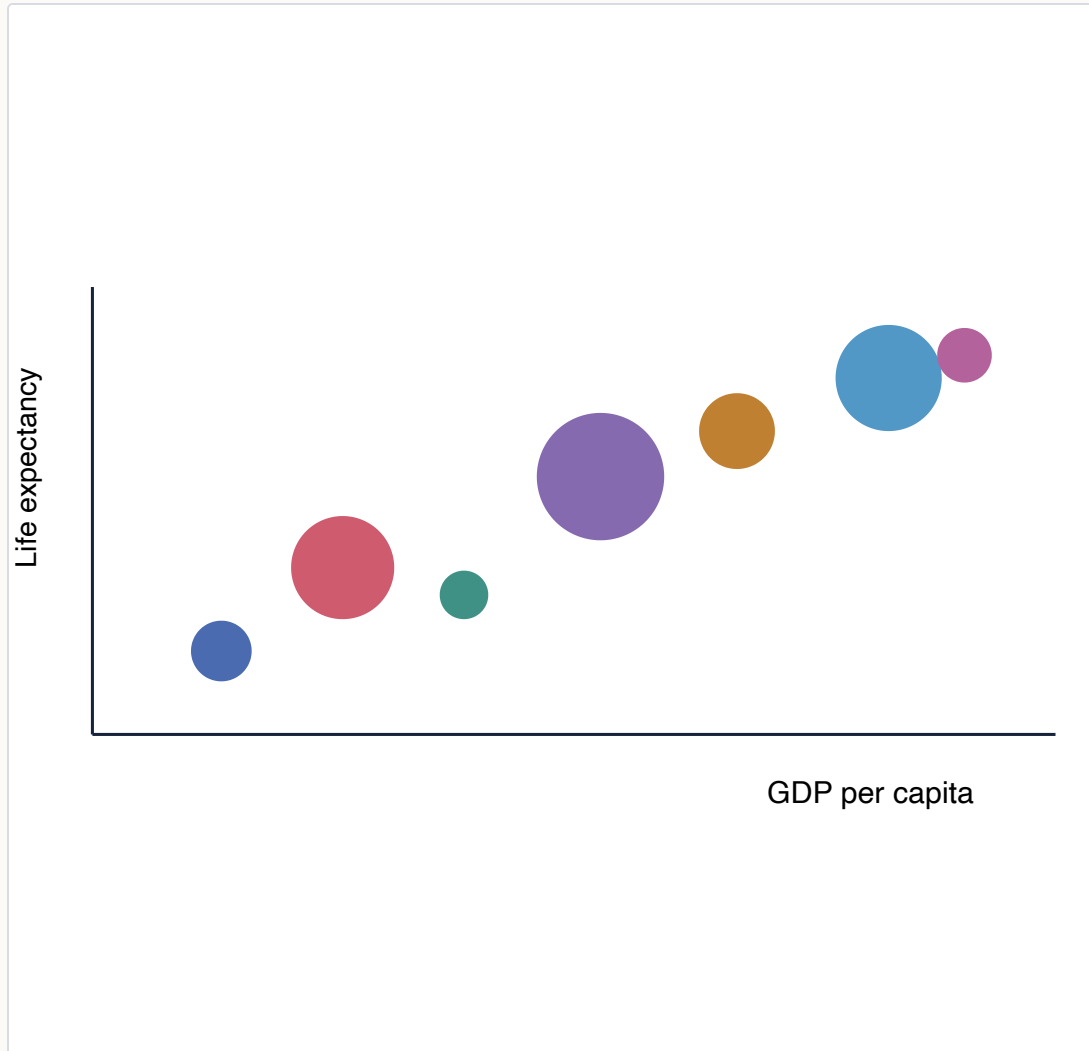
Risk

Target

Forecast

Visual emphasis works only when most elements remain visually quiet.

Evaluate the encoding in the context of its task



TASK

Examine the relationship between GDP and life expectancy across countries.

Expressiveness

GDP and life expectancy are quantitative attributes. Both use ordered position channels.

Effectiveness

The two attributes central to the task receive x and y position, supporting accurate comparison of the overall pattern.

Secondary encodings

Population uses area and continent uses hue. They add context, but the primary task does not require them.

More encoded attributes add information—and cost

Questions to ask

- Can the continent hues be distinguished?
- Do large population circles obscure other points?
- Does size interfere with hue?
- Does overlap prevent accurate reading?
- Is population needed for the stated task?

The cost of another encoding

+ information

but also

+ visual complexity

+ channel competition

+ perceptual load

Encode what matters for the task—not everything available in the dataset.

From a possible encoding to a good encoding

Expressiveness	Does the encoding preserve the meaning and relevant relationships of the data?
Effectiveness	Do task-important attributes receive perceptually strong, suitable channels?
Discriminability	Can viewers reliably distinguish the required values or categories?
Separability	Can multiple channels be read with limited interference?
Attention	Does emphasis direct viewers toward what matters without creating competition?

Lecture 3 constructs an encoding. Lecture 4 evaluates and improves it.